OFFLOAD · AGENT CORE

AGENT CORE MO: THE EVIDENCE LAYER

PR #1722, draft, 2026-11-03

CONTEXT

WHY FIRST

Today the Python agent's wire is pinned only by tests that mock the client and by Python tests that die with the service. Nothing fails if a port changes the bytes the chat client parses.

M0 adds the evidence the TypeScript port is measured against, at $0, with nothing runtime mounted in the server.

SCOPE

WHAT LANDED

Wire schemas

Strict zod wire schemas for both chat grammars and the staged card (56 exported objects, copied field by field from the Python emitters).

Citation parser

The citation-line parser moved to @offload/shared with ChatBot importing it (behavior-neutral).

Comparison kit

The QA script format and cursor, operation fixtures, a normalizer, golden and parity comparators, fixture hygiene, a port ledger.

Capture

A loopback-only capture runner. 15 goldens captured from the Python harness with their meta.

EVIDENCE

THE GOLDENS

```
__THREAD_ID__:4f709ad3-9cf3-49ee-99a9-f8822b7a95f2
Reading the document.
__STAGED__:{"set_id": "4ae3f373-811f-4ea0-bd03-094c16cfc219", "thread_id": "4f709ad3-9cf3-49ee-99a9-f8822b7a95f2", "kind": "create",
Read load 5172908 from the document into a proposed new job. Check the card and press Create.
```

The lwai-d-r1-t1 golden, all three lines, the card line cut at 140 characters (synthetic data). Captured twice from a clean environment, byte-equal both runs. 15 captured, 0 hand-authored.

PROOF

GATES

```
OFF-3954 M0 gates at 1600984a3 (origin/main ef6e64a46), 2026-11-03, Node v24.21.0
13 suites, 120 cases, 0 failures; every red edit of section 4 seen red and reverted
PASS G0.1 agent-wire.spec.ts
     Tests:   22 passed, 22 total
PASS G0.2 citation-lines.spec.ts
     Tests:   6 passed, 6 total
PASS G0.3 ChatBot.citation.spec.ts
     Tests:   2 passed, 2 total
PASS G0.4 script-cursor.spec.ts
     Tests:   10 passed, 10 total
PASS G0.5 fixture-hygiene.spec.ts
     Tests:   9 passed, 9 total
PASS G0.6 golden-compare.spec.ts
     Tests:   14 passed, 14 total
PASS G0.7 parity-compare.spec.ts
     Tests:   13 passed, 13 total
PASS G0.8 port-ledger.spec.ts
     Tests:   7 passed, 7 total
PASS G0.9 agentStreamGolden.spec.ts
     Tests:   14 passed, 14 total
PASS G0.10 harness-selftest.integration.spec.ts
     Tests:   5 passed, 5 total
PASS G0.11 capture-runner.spec.ts
     Tests:   6 passed, 6 total
PASS G0.12 operation-fixtures.spec.ts
     Tests:   9 passed, 9 total
PASS G0.13 chatFixtureTripwire.spec.ts
     Tests:   3 passed, 3 total
```

13 suites, 120 cases, 0 failures at the PR head. Every red edit of the ticket's gate table seen red and reverted.

PROOF

THE CONTRACT

```
The wire contract this PR pins: strict zod exports of the three new files (56 objects, 2 strictness checks each)
+ agent-staged-wire.ts: 39 exports
+   AGENT_STAGED_STATES
+   AGENT_STAGED_SECTIONS
+   AGENT_STAGED_SUGGESTIONS_MAX
+   AGENT_STAGED_NOT_FOUND
+   AGENT_STAGED_BUSY
+   agentStagedSourceWireSchema
+   agentStagedValueWireSchema
+   ... 32 more
+ agent-stream-wire.ts: 21 exports
+   AGENT_SENTINEL
+   AGENT_CLIENT_CAPS_HEADER
+   AGENT_STAGED_CAP
+   AGENT_STAGED_NOTICE
+   AGENT_STREAM_FAILED_TEXT
+   AGENT_CONTEXT_BLOCKED_LINE
+   AGENT_KEEPALIVE_SECONDS
+   ... 14 more
+ permit-chat-wire.ts: 26 exports
+   PERMIT_CHAT_BOTS
+   PERMIT_CHAT_SCRA_BOTS
+   PERMIT_CHAT_EVENTS
+   PERMIT_CHAT_ERROR_MESSAGE
+   PERMIT_CHAT_MAX_ITERATIONS
+   PERMIT_CHAT_EXHAUSTED_TEXT
+   PERMIT_CHAT_SESSIONS_LIMIT_MAX
+   ... 19 more
- libs/client/src/services/agentStaged.ts: TypeScript types only, no runtime check, missing keys Python writes
-   StagedSource
-   StagedEnd
-   StagedStop
-   StagedCarrier
-   StagedCommodity
-   StagedScalar
```

Strict exports on the left. The client's loose types on the right, which is what the port replaces.

TOTALS

NUMBERS

Measure	Value
Python tests collected	976
Ledger rows	976, non-pending 29
Strictness spec	265 to 377 cases
Goldens	15 (30 files)
Files in the PR	109
Base	`ef6e64a46`
Cost	$0

NEXT

NEXT

M1 builds the core (the loop, the provider seam, the spend guard, receipts, sessions, the stream writers) behind the schemas and goldens M0 pinned.

Promotion of #1722 out of draft waits on the owner.